Lesson plan · for the teacher

Teaching cyber safety across the school

Cyber security · Level 4 · Lesson 4

Time
40 min

Materials
The alphabet on the board, with a second row shifted by three letters; Scrap paper for each pair; One longer secret message, written out for the crack challenge

Learning goals

- Plan cyber safety little and often, across subjects and year groups.
- Match activities to pupils' ages.
- Respond safely when a pupil reports online harm, and involve parents.

Activity: Secret codes: hide a message, then crack one

1. Warm-up (5 min): Ask: you pass a note in class and the wrong person picks it up. How could you make sure they can't read it? Collect two or three ideas.
2. Teach (8 min): Show Caesar's cipher on the board: the alphabet with a second row shifted three letters. Encode one word together. Explain: the shift is the key; whoever has the key can read it.
3. Code (12 min): In pairs, each pupil hides a short message (no names, nothing private) with a shift they choose, writes the shift on a separate slip, and swaps. Partners decode with the key.
4. Crack (10 min): Give the class a longer message without the key. Pupils count which letter appears most and guess it stands for the most common letter in the language. Tell them Al-Kindi described this in Baghdad more than a thousand years ago.
5. Wrap-up (5 min): Ask: why was the cipher easy to crack? (Few keys, and letter counts give it away.) Explain that chat apps use keys so many that trying them all would take longer than the age of the universe.

Write the alphabet on the board with a second row shifted by three. In Arabic classes, use the 28 letters in alphabetical order. Prepare one longer message for the crack challenge, at least three lines, so the letter counts work. As a full lesson this takes 40 minutes; for a ten-minute slot in another subject, do a short Teach and one Code round.

Discussion

- Which subject could add ten minutes of cyber safety this term?
- What do younger pupils need to hear, and what do older ones need?
- Who is our safeguarding lead, and how do we reach them straight away?

Exit ticket

- One subject, one year group, and the ten-minute activity I will plan.

Try it at school tomorrow
Ten minutes of secret codes: pupils hide a short message by shifting each letter three places, swap with a partner, and decode it. Then give them a longer message without the key and let them crack it by counting the most common letter.

Practice questions · for you

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1 **A pupil tells you someone online is threatening to share a photo of her unless she pays. What do you do first?**

- ◯ Ask her to send you the photo, as evidence
- ◯ Tell her to delete the chat and block him, so it's over
- ◯ Listen calmly, don't blame, and tell the safeguarding lead straight away

2 **Match each subject to a ten-minute cyber activity.**

Maths	• • • •	Find the persuasive words in a scam message
Languages	• • • •	Discuss privacy as a right
Science	• • • •	Crack a cipher by counting letters
Social studies	• • • •	Show how a message travels across a network

3 **Pupils under 13 don't need cyber safety lessons, because most apps are for ages 13 and over.**

◯ True ◯ False

4 **Which belong in a cyber safety lesson for seven-year-olds? Choose all that apply.**

- ☐ Set up an authenticator app for your email
- ☐ Compare the security codes of encryption keys
- ☐ Keep your name, address and school private online
- ☐ Ask a trusted adult before you click or share
- ☐ Be kind online, as you are in the playground

5 **A pupil tells you about online threats. Put your response in order.**

- ☐ Tell them they were right to tell you, without blame
- ☐ Check in with the pupil, as the procedure says
- ☐ Listen calmly and believe them
- ☐ Note what they said, and ask them to keep the messages, never to send you images
- ☐ Tell the safeguarding lead straight away

✎ **Which subject you teach could carry ten minutes of cyber safety this term? What would those ten minutes be?**

Check your answers

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1 **Listen calmly, don't blame, and tell the safeguarding lead straight away**

Calm listening keeps her talking, and the safeguarding lead knows the procedure.

2 **Maths — Crack a cipher by counting letters**
Languages — Find the persuasive words in a scam message
Science — Show how a message travels across a network
Social studies — Discuss privacy as a right

Each activity uses the subject's own skills, so it strengthens the lesson instead of interrupting it, and pupils meet cyber ideas in many places.

3 **False**

Many younger children are online through games, videos and a parent's phone.

4 **Keep your name, address and school private online; Ask a trusted adult before you click or share; Be kind online, as you are in the playground**

Young children need a few simple habits they can act on.

5 **Top to bottom: 2, 5, 1, 3, 4**

Listening and reassurance keep the pupil talking; your notes and the pupil's own messages keep the evidence without spreading the images; the safeguarding lead brings the right help, and following up shows the pupil they were taken seriously.